C2 metrics: definitions, measured results and interpretation

Printable edition. Use the Contents links or PDF bookmarks to navigate. Long code lines wrap for printing; copy executable commands from the Markdown guide. Source-file links refer to the local repository; keep the three PDFs together for related-guide links.

Each local file link is followed by its full path as plain text. A final `:number` identifies the source line; in VS Code, open the file with `Ctrl+P` and jump to that line with `Ctrl+G`.

Snapshot: 15 October 2026. These tables come from saved non-test outputs, not new training or test evaluation. Display values are rounded to six decimal places; original CSV/JSON files preserve precision. All recovery errors below use research coordinates z unless stated otherwise. Read with the component guide (path: `D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/COMPONENT_STUDY_GUIDE.md`).

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1. Which kind of number am I looking at?

Number	Simple meaning	Available for a new unknown clip?
Predicted z / Vital n	The model's proposed knob setting	Yes
Parameter error, MAE, RMSE, R2	Difference from known original settings	Only if original settings are supplied independently
U	Ensemble's predictive variance, in z squared	Yes, from trained models
R	Measured difference between input and predicted sound	Yes, after rendering
S_j	Audio change when parameter j moves a little	Yes, after perturbation renders
q_j	Fitted probability of original-parameter recovery within tolerance	Yes, with calibration limitations
Brier, ECE, failure AUROC/AP, AURC	Population evaluation of reliability outputs	Need a labelled evaluation set
Peak/RMS/DC/hash	Signal quality or integrity checks	Yes; they do not measure parameter accuracy

There is no validated universal low/medium/high score scale, perceptual R cutoff or operational Accept/Approximate/Abstain policy yet.

2. Recovery metrics in plain language and equations

Let `e_ij = predicted_z_ij - original_z_ij`, for clip i and parameter j.

Metric	Formula for one parameter	Explanation
MAE	mean(abs(e))	Average distance from the correct knob position
MSE	mean(e^2)	Squared-error training/evaluation objective; large misses matter more
RMSE	sqrt(mean(e^2))	Squared-error measure returned to the original coordinate units
R2	1 - sum(e^2)/sum((true_z-mean(true_z))^2)	Improvement over predicting this evaluation set's mean; 1 is perfect, 0 matches that reference, negative is worse

R2 is undefined if the target is constant. It is not a probability or a percentage of correct knobs. The saved training-mean predictor is a separate practical baseline; its constant comes from training, whereas the R2 denominator uses the evaluated target mean. R2 is unchanged by an affine coordinate rescaling; MAE/RMSE are not.

An MAE .05 means 5% of a parameter's restricted interval, not 5% of full physical frequency/time or a perceptual threshold. Macro MAE gives each of 13 parameters equal weight. Averaging 13 RMSEs is not the same as taking the square root of the all-parameter MSE; say which aggregation you use.

Code: baseline.regression_metrics (path: `D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/v4/baseline.py:53`).

3. Single CNN and ensemble: the same 4,096 inverse-validation clips

Parameter	Single MAE	Single RMSE	Single R2	Ensemble MAE	Ensemble RMSE	Ensemble R2
wave_frame	0.078432	0.106992	0.861098	0.066230	0.091645	0.898087
bend_amount	0.216818	0.262532	0.165682	0.241761	0.280462	0.047828
filter_cutoff	0.056082	0.081588	0.919057	0.044266	0.067620	0.944400
filter_resonance	0.149575	0.191556	0.552953	0.156537	0.198985	0.517608
filter_drive	0.180036	0.221752	0.404213	0.215541	0.254445	0.215587
amp_attack	0.040772	0.053499	0.965886	0.015749	0.026980	0.991324
amp_decay	0.090675	0.131770	0.790556	0.076709	0.126803	0.806047
amp_sustain	0.049487	0.066501	0.947826	0.031470	0.048848	0.971849
amp_release	0.025746	0.035617	0.984816	0.003329	0.005081	0.999691
filter_env_amount	0.168136	0.210775	0.464376	0.186703	0.230561	0.359095
filter_env_attack	0.171434	0.213378	0.457955	0.196728	0.238382	0.323473
filter_env_decay	0.223305	0.266838	0.158404	0.240880	0.281031	0.066491
filter_env_sustain	0.188961	0.233032	0.346583	0.187984	0.233534	0.343768

Single CNN macro MAE: **.126112**; ensemble macro MAE: **.127991**. The ensemble improves some parameters but has slightly worse average MAE. Single CNN MSE is .031867 versus .083213 for the training-mean baseline; ensemble MSE is .035284. The two models use different training objectives, so differences do not isolate ensemble averaging alone.

Sources: single CNN metrics (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/baseline_full_v1/validation_metrics.csv`), ensemble metrics (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/ensemble_full_v1/parameter_metrics.csv`), training-mean baseline (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/baseline_full_v1/mean_baseline_metrics.csv`). Each recovery CSV also includes Vital-normalized coordinate metrics.

4. What is a correctness label?

`y_ij = 1[abs(predicted_z_ij - original_z_ij) <= tau]`. Primary tau=.05; secondary tau=.025/.10. The boundary is inclusive. The model estimates q=P(y=1 | measured features), not P(perceptually identical audio).

The following is a **different** set: the 512 reliability-validation clips. It must not be substituted for the 4,096-clip inverse-validation result.

Parameter	MAE z	RMSE z	R2	Successes / 512	Failures
wave_frame	0.066408	0.093852	0.887424	257	255
bend_amount	0.237756	0.278356	0.025322	59	453
filter_cutoff	0.043168	0.066148	0.947494	369	143
filter_resonance	0.161219	0.205101	0.496555	112	400
filter_drive	0.220873	0.261189	0.199929	64	448
amp_attack	0.014941	0.026328	0.991552	474	38
amp_decay	0.073794	0.119388	0.838172	306	206
amp_sustain	0.031281	0.046633	0.973734	422	90
amp_release	0.003575	0.007047	0.999396	510	2
filter_env_amount	0.187707	0.230744	0.382301	81	431
filter_env_attack	0.181858	0.223940	0.338150	86	426
filter_env_decay	0.236488	0.277070	0.062193	50	462
filter_env_sustain	0.202163	0.250733	0.303487	88	424

Failure counts matter. Amplitude release has only two failures, so its ranking metrics can change greatly if just one example changes. Bend and filter-envelope decay have many failures despite some patches reconstructing reasonably well.

5. Probability, ranking and selective metrics

Metric	Simple explanation	Exact policy here
Brier	Squared error of a probability against a 0/1 outcome	mean((q-y)^2); lower is better; primary probability score
Reliability diagram	Compare what q says with how often predictions succeed	Plot bin mean q against observed success; diagonal is agreement
ECE	Average gap between stated probability and observed frequency	10 equal-frequency bins, weighted absolute gap; lower is better
Failure AUROC	Does a failing estimate usually get a higher failure score than a successful one?	Failure=1-y; score=1-q, or raw U; average-rank ties; higher is better
Failure AP	Precision of detected failures as more are included	Noninterpolated average precision; score ties grouped; higher is better
Coverage	Fraction of parameter predictions kept	k/n after sorting by q descending, or U ascending
Risk at coverage	Fraction of kept estimates outside tolerance	Cumulative failures/k; lower is better
AURC	Average retained failure risk along the coverage curve	Arithmetic mean over prefixes k=1..n; lower is better
MAE at coverage	Parameter error among estimates kept at the same rate	Cumulative absolute z error/k; lower is better

Brier evaluates probability quality, not calibration in isolation. A constant can score well on an imbalanced outcome; this is why g0, class counts and ranking are reported too. ECE depends on binning and is not proof of perfect calibration. AUROC is .5 for a constant score with both classes present. Random-score AP is related to failure prevalence, so compare AP within the same parameter/outcome, not against .5 universally. The implementation reports **AP**, not trapezoidal PR-AUC; do not silently rename the statistic.

Single-class AUROC/AP are undefined. q sorting is stable in preassigned diagnostic order. Equal-q ties may be split between ECE bins. A constant predictor's risk curve follows that fixed tie order, not learned discrimination. At nominal 10% coverage, ceil(512*.1)=52 predictions are kept; actual coverage is 52/512. At 50%, exactly 256 predictions per parameter are kept.

These are parameter-level selections. Different parameters can retain different audio IDs; this does not define a whole-patch acceptance rule.

The exact calculation:

c2_estimation/v4/calibration.py:111 (path: `D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/v4/calibration.py:111`)

```
def probability_metrics(y,q,error):
    if not np.isfinite(q).all() or np.any((q<0)|(q>1)):
        raise ValueError("Invalid correctness probabilities")
    ordered = np.argsort(q,kind="stable")
```

```
    bins = []
    for k,ids in enumerate(np.array_split(ordered,min(10,len(y)))):
        bins.append({"bin":k,"count":len(ids),"mean_q":float(q[ids].mean()),"observed_success":float(y[ids].mean()),
                     "min_q":float(q[ids].min()),"max_q":float(q[ids].max())})
    ece = sum(b["count"]*abs(b["mean_q"]-b["observed_success"]) for b in bins)/len(y)
    coverage,risk,mae = risk_curve(y,q,error)
    result = {"brier":float(np.square(q-y).mean()),"ece_10_equal_frequency":float(ece),
              "aurc_discrete":float(risk.mean()),"success_count":int(y.sum()),"failure_count":int((1-y).sum()),
              **ranking_metrics(1-y,1-q)}
    for c in COVERAGES:
        i = max(1,int(np.ceil(len(y)*c)))-1
        result[f"risk_at_coverage_{c:g}"] = float(risk[i])
        result[f"mae_z_at_coverage_{c:g}"] = float(mae[i])
    return result,bins,(coverage,risk,mae)
```

Related code: ranking_metrics (path: `D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/v4/calibration.py:85`), risk_curve (path: `D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/v4/calibration.py:102`).

6. All five probability models: macro development results

Model	Macro Brier	Macro ECE	Macro AURC
g0	0.138242	0.040681	0.568125
g(U)	0.116267	0.037460	0.462585
g(U,S)	0.115908	0.036614	0.459520
g(U,R)	0.115572	0.035921	0.462154
g(U,S,R)	0.115385	0.035282	0.459777

These are equal-weight macro averages over 13 per-control metrics. g(U,S) has slightly lower AURC than the full model, while the full model has slightly lower Brier/ECE. Raw U has no probability Brier/ECE because variance is not q. Reliability-validation selected C, so these are development results.

7. Per-parameter Brier for every probability ablation

Parameter	g0	g(U)	g(U,S)	g(U,R)	g(U,S,R)
wave_frame	0.250798	0.195597	0.189182	0.187582	0.183839
bend_amount	0.102051	0.075224	0.074995	0.074836	0.074752
filter_cutoff	0.201672	0.166640	0.166705	0.166238	0.166348
filter_resonance	0.170927	0.139412	0.139774	0.139140	0.139402
filter_drive	0.109394	0.107745	0.107746	0.107286	0.107240
amp_attack	0.068719	0.048988	0.049959	0.049244	0.049713
amp_decay	0.240589	0.170184	0.169817	0.169991	0.169662
amp_sustain	0.145126	0.118832	0.119663	0.118487	0.119251
amp_release	0.003900	0.002918	0.003061	0.002903	0.003020
filter_env_amount	0.133177	0.128986	0.128839	0.129663	0.129552
filter_env_attack	0.140100	0.135815	0.136123	0.135901	0.136226
filter_env_decay	0.088188	0.088122	0.088013	0.088181	0.088088
filter_env_sustain	0.142508	0.133004	0.132925	0.132983	0.132908

Lower is better. Wave frame contributes much of the improvement over U alone. This table exposes small regressions as well as improvements; the full model does not improve every parameter.

8. Full U+S+R reliability results for all parameters

Parameter	Brier	ECE	Failure AUROC	Failure AP	AURC
wave_frame	0.183839	0.063845	0.796582	0.756097	0.266263
bend_amount	0.074752	0.025909	0.818274	0.968973	0.731550
filter_cutoff	0.166348	0.039445	0.761669	0.555094	0.132704
filter_resonance	0.139402	0.042601	0.768326	0.909925	0.603304
filter_drive	0.107240	0.036229	0.607910	0.905919	0.822321
amp_attack	0.049713	0.016051	0.938763	0.452329	0.007588
amp_decay	0.169662	0.034534	0.816597	0.755776	0.188760
amp_sustain	0.119251	0.044865	0.819510	0.449437	0.055260
amp_release	0.003020	0.003348	0.790196	0.504630	0.001073
filter_env_amount	0.129552	0.042942	0.629315	0.901057	0.784369
filter_env_attack	0.136226	0.043839	0.617234	0.885317	0.769311
filter_env_decay	0.088088	0.019180	0.542121	0.914386	0.888395
filter_env_sustain	0.132908	0.045873	0.693075	0.899882	0.726209

ECE uses the declared ten-bin policy. AUROC/AP target failures. An impressive amplitude-release ranking number is fragile with only two failures; a high AP for bend partly reflects that failures are common. Check the counts in section 4.

9. Raw uncertainty ranking before logistic calibration

Parameter	Failure AUROC	Failure AP	AURC
wave_frame	0.775738	0.762988	0.303827
bend_amount	0.815991	0.969646	0.733393
filter_cutoff	0.760475	0.551599	0.132233
filter_resonance	0.764241	0.906549	0.603546
filter_drive	0.575718	0.898192	0.831529
amp_attack	0.940207	0.464895	0.007457
amp_decay	0.814868	0.753220	0.188112
amp_sustain	0.823091	0.449212	0.054224
amp_release	0.927451	0.513158	0.000317
filter_env_amount	0.644267	0.903533	0.777903
filter_env_attack	0.624986	0.887161	0.766565
filter_env_decay	0.534589	0.915550	0.887312
filter_env_sustain	0.688974	0.895382	0.727180

Raw U ranks likely failures directly. g(U) is a learned monotonic logistic transformation of log U, whose fitted sign can differ by parameter; therefore do not assume their rankings must always be identical. Neither is guaranteed to identify recordings outside the training domain.

10. Same 50% coverage: retain 256 settings per parameter

Parameter	Raw-U risk	U risk	U+S+R risk	Raw-U MAE	U MAE	U+S+R MAE
wave_frame	0.289062	0.289062	0.257812	0.037209	0.037209	0.037498
bend_amount	0.796875	0.796875	0.804688	0.132718	0.132718	0.137616
filter_cutoff	0.144531	0.144531	0.136719	0.025886	0.025886	0.025240
filter_resonance	0.656250	0.656250	0.656250	0.108292	0.108292	0.111858
filter_drive	0.855469	0.855469	0.847656	0.197020	0.197020	0.194547
amp_attack	0.000000	0.000000	0.000000	0.005521	0.005521	0.006015
amp_decay	0.164062	0.164062	0.160156	0.028808	0.028808	0.028426
amp_sustain	0.039062	0.039062	0.046875	0.017603	0.017603	0.017931
amp_release	0.000000	0.000000	0.000000	0.002521	0.002521	0.002528
filter_env_amount	0.792969	0.792969	0.804688	0.142528	0.142528	0.155169
filter_env_attack	0.800781	0.800781	0.792969	0.150692	0.150692	0.149225
filter_env_decay	0.898438	0.898438	0.890625	0.228176	0.228176	0.238795
filter_env_sustain	0.761719	0.761719	0.761719	0.153296	0.153296	0.157632

Compare methods at equal retained count; a method cannot gain credit merely by keeping fewer predictions. All variants and nominal coverages .1/.25/.5/.75/1 are preserved in parameter_metrics.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/parameter_metrics.csv`), raw_uncertainty_ranking.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/raw_uncertainty_ranking.csv`) and matched_coverage_audit.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/results_audit_v1/matched_coverage_audit.csv`). The full prefix curves are in risk_coverage.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/risk_coverage.csv`).

11. Different definitions of correct: secondary tolerance analysis

Tolerance z	g0	g(U)	g(U,S)	g(U,R)	g(U,S,R)
0.025000	0.122540	0.103842	0.103434	0.103254	0.102946
0.050000	0.138242	0.116267	0.115908	0.115572	0.115385
0.100000	0.146619	0.122370	0.122377	0.122240	0.122300

Each row fits its own calibrators for that correctness event, using the same predictions and .02 diagnostic renders. Do not treat a smaller Brier at one tau as proof that that tau is better: event prevalence changes with tolerance. Primary tau remains .05. Sources are the `summary.json` files in `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.025`, `calibration_tau_0.05` and `calibration_tau_0.1`.

12. Paired uncertainty around Brier differences

Comparison: full minus reference	Point difference	Paired 95% percentile interval
g(U), all 13	-.000882	[-.001684, -.000089]
g(U,R), all 13	-.000187	[-.000757, .000400]
g(U), exclude wave frame, exploratory	+.000024	[-.000422, .000486]
g(U,R), exclude wave frame, exploratory	+.000109	[-.000291, .000517]

2,000 paired whole-audio resamples keep each clip's 13 controls together. The conditional interval versus U+R includes zero. Thus additional benefit from S is not established in this development analysis. The exploratory exclusion is not the primary endpoint; no per-control multiple-comparison correction was applied. Intervals condition on the fitted models and development selection and do not include uncertainty from retraining.

Exact source: macro_paired_comparisons.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/results_audit_v1/macro_paired_comparisons.csv`), per_parameter_paired_comparisons.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/results_audit_v1/per_parameter_paired_comparisons.csv`).

13. Reconstruction residual R and sensitivity S

At each FFT resolution:

```
SC = ||A-B||F / (0.5*(||A||F + ||B||F) + 1e-12)
LM = mean(abs(ln(max(A,1e-4)) - ln(max(B,1e-4))))
D = mean_over_512_1024_2048(SC + LM)
R = D(target, predicted_audio)
S_j = D(minus_audio_j, plus_audio_j) / actual_readback_span_z_j
```

R and S are nonnegative numerical distances/activity scores. They do not have a validated good/bad cutoff and cannot be converted to percentages of similarity. The per-resolution components are retained in case JSON records.

Descriptive R distribution from already-saved non-test diagnostic files:

Partition	Clips	Mean R	Median R	90th percentile R	Min R	Max R
calibration	2048	0.313639	0.289749	0.523543	0.037167	0.996517
reliability-validation	512	0.319013	0.301779	0.524795	0.043359	0.872882

These are absolute residual summaries, not an audio-reconstruction comparison against a population-wide midpoint baseline or a listening study. The known-true roundtrip and midpoint comparisons elsewhere concern selected worked examples.

14. Perturbation-step agreement on 256 calibration clips

Parameter	rho .01/.02	rho .02/.05	p95 abs log1p difference .02/.05
wave_frame	0.976079	0.924167	0.689971
bend_amount	0.980507	0.938325	0.693299
filter_cutoff	0.999386	0.995610	0.215802
filter_resonance	0.999999	0.999958	0.004595
filter_drive	0.999999	0.999972	0.001051
amp_attack	0.998069	0.992270	0.404696
amp_decay	0.999987	0.999964	0.008606
amp_sustain	0.999992	0.999824	0.007125
amp_release	0.999971	0.999812	0.008518
filter_env_amount	0.999999	0.999929	0.007522
filter_env_attack	0.999621	0.998441	0.078085
filter_env_decay	0.999914	0.999752	0.006365
filter_env_sustain	0.999996	0.999917	0.003133

Spearman compares rank ordering, not magnitude. The p95 column measures the 95th percentile of abs(log(1+S_first)-log(1+S_second)), not a percentage or dB value. Near-perfect ranks can coexist with changing magnitudes, especially amp attack. Wave frame/bend are more scale-dependent. Keep .02; no calibrator was refitted.

All pairs, zero counts, group counts and modes are in step_size_agreement.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/step_stability_v1/step_size_agreement.csv`). The complete 9,984 measurements are in step_size_measurements.csv (path: `D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/step_stability_v1/step_size_measurements.csv`). Tiny boundary groups and undefined constant/empty correlations must be interpreted cautiously.

15. Audio QA numbers are different again

Peak=max(abs(audio)); RMS=sqrt(mean(audio^2)); RMS dBFS=20*log10(RMS); DC=mean(audio). Full generation flags silence at RMS below -60 dBFS, clipping at peak>=1 and a DC warning at abs(mean)>=.005 or abs(mean)/RMS>=.20. The completed conditioned corpus has zero flags under those rules. This does not say all raw audio had zero DC.

Exact hash equality, repeat rendering and known-parameter roundtrip are implementation checks. They do not measure perceptual quality, correctness probability or proof of unique parameter recovery.

16. Where are all remaining numbers?

Evidence	Exact file
Member seeds/epochs/checkpoint hashes/variance summaries	ensemble summary (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/ensemble_full_v1/summary.json)
Epoch-by-epoch losses and errors	member 0 history (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/ensemble_full_v1/member_00/training_history.csv), equivalent files under member_01/member_02
Selected C, fitted weights, transforms and calibration class counts	calibrators.json (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/calibrators.json)
Every candidate C validation score	candidate_C_scores.csv (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/candidate_C_scores.csv)
Reliability bin counts and observed frequencies	reliability_bins.csv (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/reliability_bins.csv)
Each clip's frozen prediction/U/S/R	diagnostics.csv (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/diagnostics_reliability-validation/diagnostics.csv)
Actual perturbation endpoints and denominators	perturbation_readbacks.csv (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/diagnostics_reliability-validation/perturbation_readbacks.csv)
Delivered q values for every model and clip	validation_probabilities.parquet (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/development_uncertainty_v1/calibration_tau_0.05/validation_probabilities.parquet)
Implementation checks and skip reasons	verification.json (path: D:/VSCode/GitHub/Synth-Inversion-RP/data/processed/c2_v4/verification_validation_viewer_v1/verification.json)
Snapshot source fingerprints	STUDY_GUIDE_EVIDENCE.json (path: D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/docs/STUDY_GUIDE_EVIDENCE.json)

The read-only collector is collect_study_evidence.py (path: D:/VSCode/GitHub/Synth-Inversion-RP/c2_estimation/docs/collect_study_evidence.py). It reads saved tables and source excerpts, checks expected dimensions/non-test flags, and prints documentation evidence. It does not rerender, retrain, refit or evaluate test audio. These documents describe this snapshot, not automatic future results.